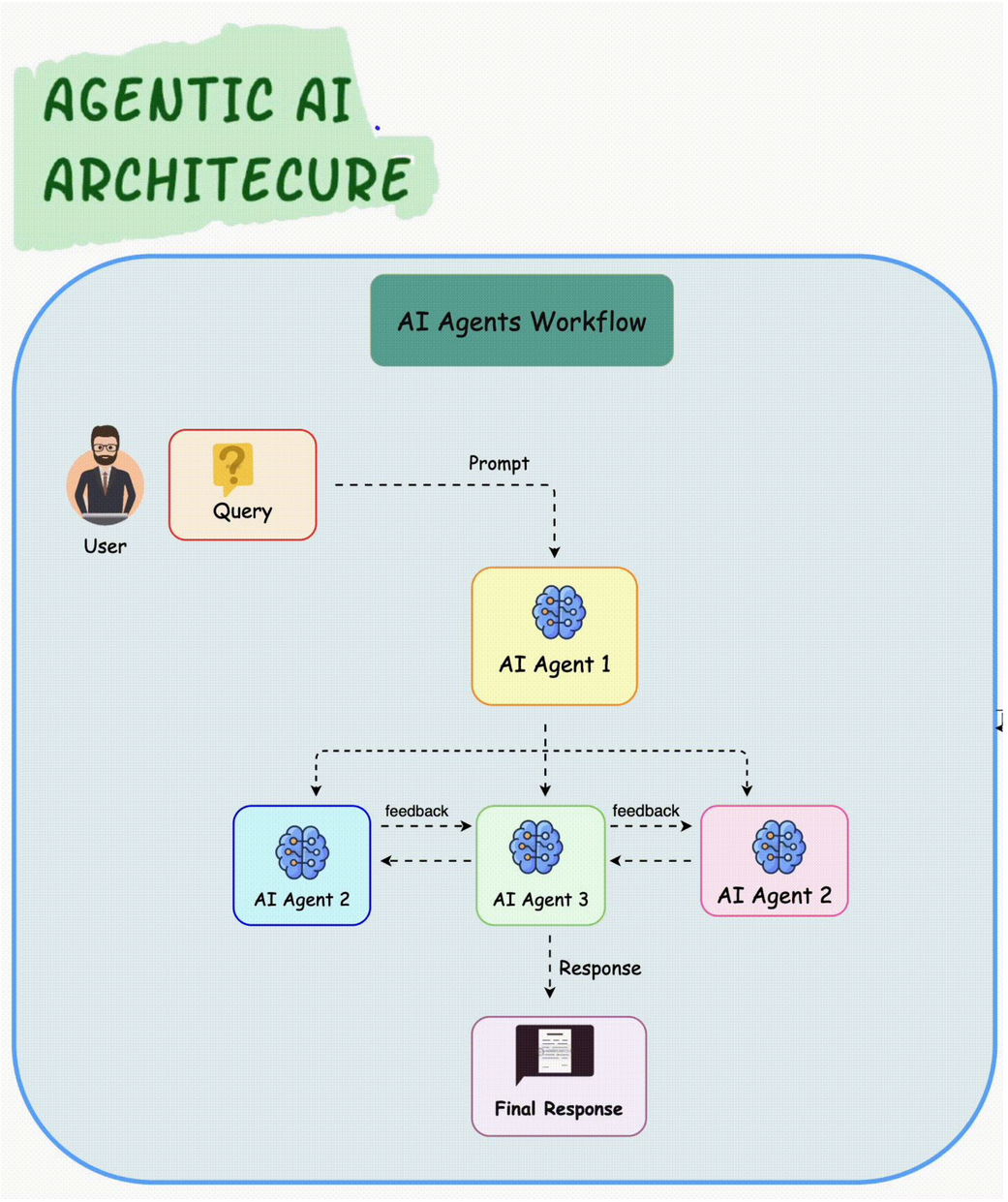
agents and tools

week 4 of natural language processing track

presentation by mehek and sania!



what is an agent????

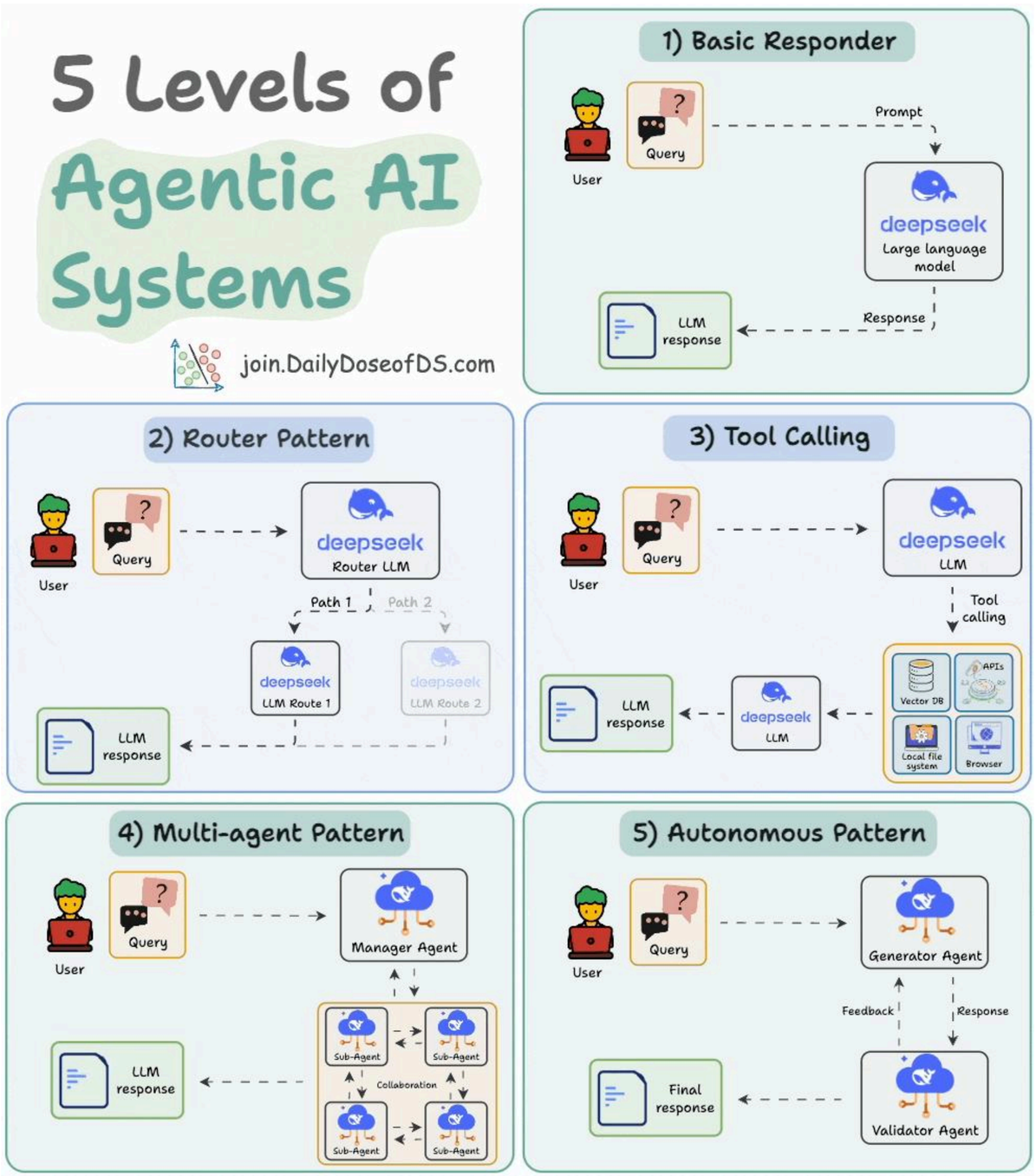
an agent is defined by its ability to act on its own to achieve a goal, rather than just generating text based on a single prompt

specifically designed to complete a task

theres levels to it:

- **basic prompting**
- **tool calling**
- **multi-agents**
- **self-improvement feedback loops**

note: level 2 in the graphic is referring to the “mixture of experts” architecture of LLMs, which we haven’t yapped about yet



task: write a college essay. [insert question] [insert max word count]

degrees of autonomy for an agent

less

- all steps predetermined
- all tool use is hard-coded
- autonomy is in text-generation only

writer agent always calls word_count() tool

semi

- agent can make some decisions
- agent chooses between tools
- all tools predefined

writer agent picks word_count() tool OR send_to_email() tool OR read_user_resume() tool

highly

- agent makes decisions independently
- can create new tools
- minimal human oversight

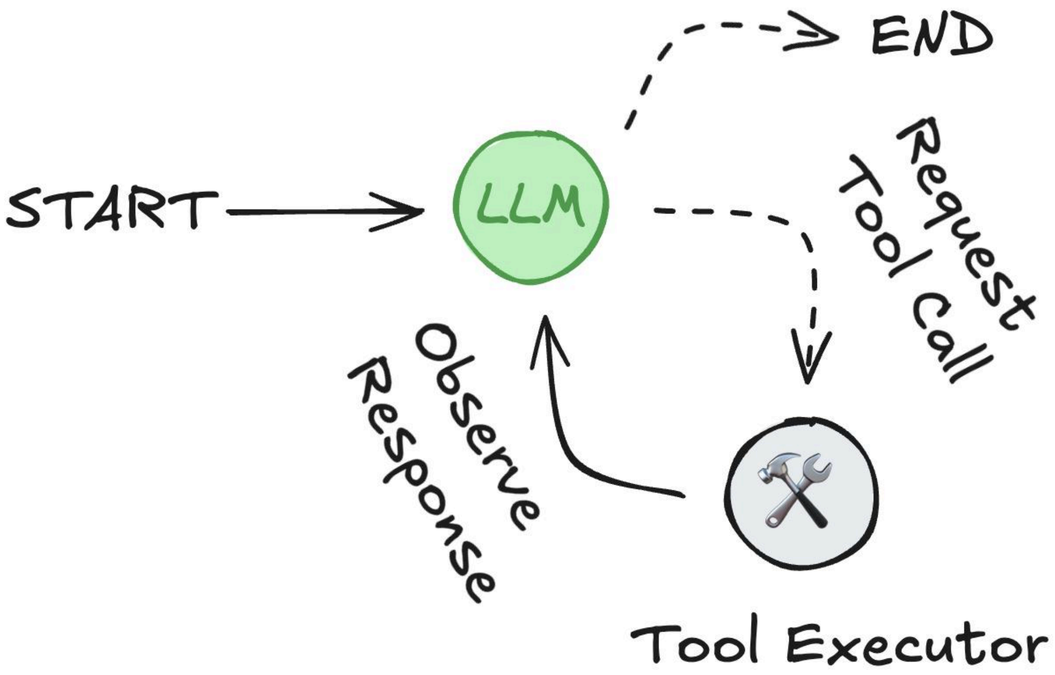
writer agent can create new tools or use pre-defined tools

editor agent evaluates the essay and sends it back to the writer for improvements

what is a tool????

it has some kind of output that the agent needs to execute the task or get more information

- a tool is a function that the agent can call to execute something
- agent calling a function is similar to a function calling another function inside it



what happens when an agent uses a tool:

1. tool options given to agent as context (functions!)
2. agent outputs string to call tool
3. fancy logic (that the model provider handles) to execute the actual tool function code
4. output from the function given back to agent

categories of tools

they all output some kind of information or execute something that the agent decides is necessary

external APIs

theres an api for basically everything

ex: api for getting random kanye west quotes https://kanye.rest/

always search up an api and research before doing a bunch of work to build something yourselfffff

user-defined

create your own custom function to use

```
def ur_mom()
    return "likes cheese"
```

raw code

code that the agent can execute in a sandbox environment

llm-sandbox

```python
from llm_sandbox import SandboxSession

# create a new sandbox session
with SandboxSession(lang="python") as session:
    result = session.run("print('Hello, World!')")
    print(result.text)  # Hello, World!
```

do not use python exec()

databases / storage

retrieve or store info

calls to other agents

talk / delegate to other agents at will

all of these tools are just code functions

metrics and evaluating agent output

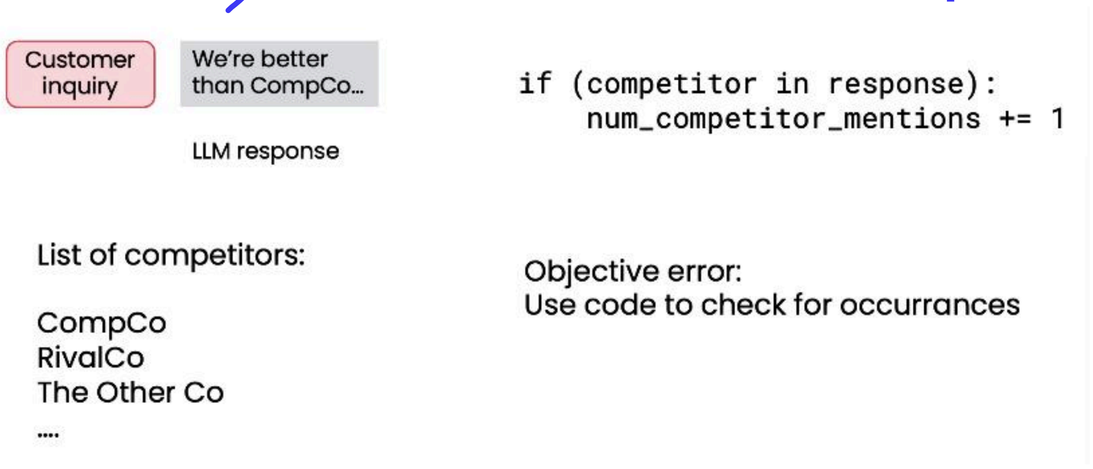
many different options to evaluate an agent's output:

- create functions that parse the string for problems
- send it to another agent to evaluate
- send it to another agent to evaluate ALONG with tools that it can use for metrics

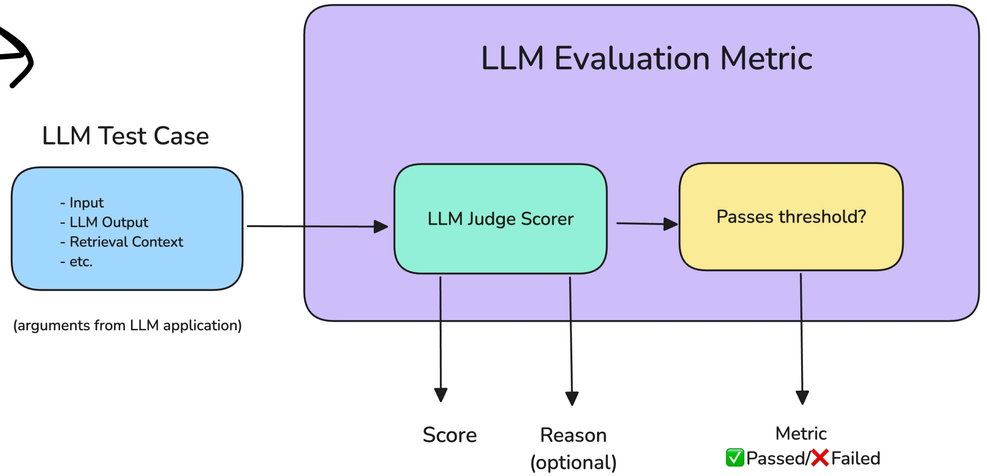
figure out what you want to do with the evaluation information:

- consider task completed
- send back to agent to refine task
- send to a different agent to continue task

for objective evaluation, based on your own observations in the output



for subjective evaluation executed by an LLM



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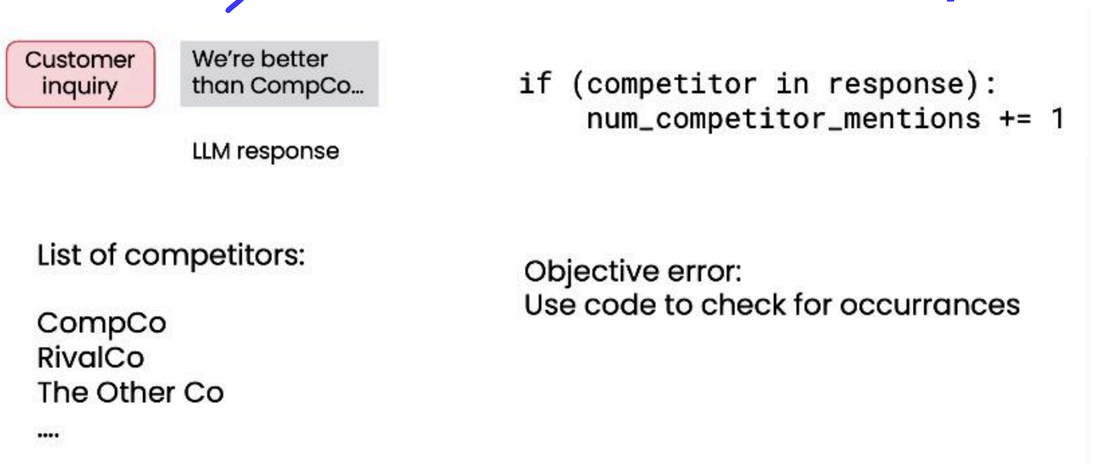
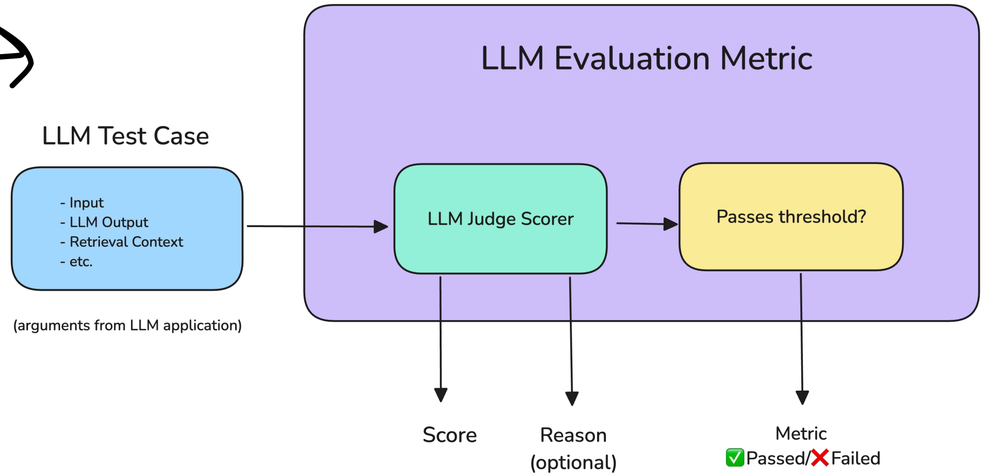
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